# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 6: **Number Play** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** Kaprekar's routine on $6321$ reaches 6174 after how many steps? 1

**(a)** 8 **(b)** 9 **(c)** 7 **(d)** 6

**2.** The Collatz sequence from $15$ needs how many steps to reach 1? 1

**(a)** 18 **(b)** 16 **(c)** 17 **(d)** 19

**3.** For every whole number $n$, $n + (n+1) + (n+2)$ is always a multiple of: 1

**(a)** 3 **(b)** 4 **(c)** 6 **(d)** 2

**4.** If $1 + 2 + 3 + \cdots + n = 406$, then $n$ is: 1

**(a)** 29 **(b)** 28 **(c)** 30 **(d)** 27

**5.** The number of reverse-and-add steps for $79$ to become a palindrome is: 1

**(a)** 5 **(b)** 6 **(c)** 7 **(d)** 8

**6.** Which expression is a multiple of 11 for every digit pair $(a, b)$? 1

**(a)** $10a + b$ **(b)** $(10a + b) - (10b + a)$ **(c)** $(10a + b) + (10b + a)$ **(d)** $a - b$

**7.** The smallest 5-digit number divisible by 9 is: 1

**(a)** 10008 **(b)** 10017 **(c)** 10007 **(d)** 10009

**8.** The largest 4-digit number with all digits different that is divisible by 10 is: 1

**(a)** 9872 **(b)** 9869 **(c)** 9870 **(d)** 9871

**9.** **Assertion (A):** The sum of a two-digit number and its reverse is divisible by 11. 1

**Reason (R):** (10a + b) + (10b + a) = 10(a + b).

Choose the correct option:

**(a)** Both A and R are true and R is the correct explanation of A.

**(b)** Both A and R are true but R is not the correct explanation of A.

**(c)** A is true but R is false.

**(d)** A is false but R is true.

**10.** **Assertion (A):** The product of two odd numbers is even. 1
**Reason (R):** The product of an even number and any whole number is even.
Choose the correct option:
**(a)** Both A and R are true and R is the correct explanation of A.
**(b)** Both A and R are true but R is not the correct explanation of A.
**(c)** A is true but R is false.
**(d)** A is false but R is true.

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Show that $n(n+1)$ is even, and verify for $n = 24$. 2

**12.** Using each of the digits $6, 4, 1, 0$ once, make the largest 4-digit number divisible by 5. 2

**13.** A two-digit number added to its reverse gives $121$ and subtracted from it gives $63$. Find the number. 2

**14.** Show that $n(n+1)$ is even, and verify for $n = 37$. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** The number $5 * 8 * 4$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

**16.** Start with $1279$ and apply Kaprekar's routine until 6174 appears. Show every step. 3

**17.** In how many ways can $27$ be written as the sum of two or more consecutive positive whole numbers? List them. 3

**OR**

The number $3 * 6 * 9$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

**18.** Start with $1327$ and apply Kaprekar's routine until 6174 appears. Show every step. 3

**OR**

In how many ways can $33$ be written as the sum of two or more consecutive positive whole numbers? List them. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study.** Arjun adds a two-digit number to its reverse. For 47 he gets $47 + 74 = 121$. 5
(i) Find the sum for 83.
(ii) Show that the sum for 47 is divisible by 11.
(iii) For a number with digits $a$ and $b$, show that the sum of the number and its reverse equals $11(a+b)$.
(iv) Without adding, use the rule to find the sum of 62 and its reverse.

**20.** Use reverse-and-add on $174$ until a palindrome is formed. Show each step, state the number of steps and the final palindrome. 5

**OR**

Find all two-digit numbers that are exactly 5 times the sum of their digits. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**